

JOHNSON MUGARRA

Research and Data Consultant, Technical Lead Consultant, Research Associate
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PROFESSIONAL SUMMARY

Research and data consultant with experience across proposal development, applied research, and field data systems. At the Center for Research and Communication, coordinated proposals and provided technical leadership for evaluations covering design, sampling, tools, data quality, analysis, and reporting. At Enimiro, built and maintained SurveyCTO to PostgreSQL data pipelines, dashboards, and farmer data systems. Conducted vanilla value-chain research on harvest timing with Makerere University, MAAIF, and partners. Earlier work includes process evaluation support at Innovations for Poverty Action, qualitative field research for USAID SITES, and research assignments at Bronkar.

CORE SKILLS

- Statistical software: Stata, R, Python, SQL, SAS, SPSS, Excel
- Research and evaluation: survey design, piloting and translation, computer-assisted interviewing, key informant interviews, focus group discussions, transcription
- Data collection and quality: questionnaires, tablet-based collection, enumerator training and supervision, validation, back-up and documentation
- Visualization and reporting: Power BI, Tableau, dashboards, donor-ready reports
- Data systems: SurveyCTO API, PostgreSQL, Google Sheets, ETL pipelines, GeoPandas
- Machine learning and AI: scikit-learn, time-series forecasting, neural networks and CNNs in TensorFlow and PyTorch, and daily AI tooling for cleaning, code, and drafting with outputs verified by hand
- Proposal development: concept note, strategy, design, methodologies and work plans, budgeting, quality assurance

WORK EXPERIENCE

Research and Data Consultant, November 2025 - September 2026

Enimiro Products Uganda Ltd, Kayunga, Uganda

- Coordinated with the data team to collect, clean, and organize operational data on farmer activities, crop production, sales, and sustainability programs.
- Maintained a Python pipeline moving SurveyCTO submissions into PostgreSQL raw, clean, and analysis tables, with scheduled exports to Google Sheets for dashboards.
- Matched GPS records to district boundaries and maintained the farmer master with merge and change logs for auditability.
- Built and maintained dashboards and reports tracking performance indicators for management and the board.
- Ran statistical and trend checks to flag risks and opportunities and translated findings into training manuals and field guidance.
- Supported crop and supply-chain research, updated survey tools, trained staff on data use, and documented methods for donor and certification reporting.

Technical Lead Consultant, January 2025 - June 2026

Center for Research and Communication (CRC), Remote

- Reviewed client requirements and evaluation criteria to shape proposal strategy and coordinated specialist inputs against deadlines.
- Wrote and edited technical sections, including methodologies and work plans, and worked with finance to align technical approaches with compliant budgets.
- Provided technical leadership for the IRC MERP endline evaluation, covering design and tools, analysis and synthesis, and integration of qualitative and quantitative findings against the results framework.
- Led quantitative inputs for IRC baseline and perception studies: sampling strategy, instrument testing, data-quality checks, and statistical analysis with visual summaries.
- Served as data and M&E lead for UNICEF behavior-change and awareness studies: tools, trackers, datasets, reach analysis, and report drafting.

Researcher / Data Scientist, Vanilla Value-Chain Research, January 2024 - June 2025

Makerere University with MAAIF, Catholic Relief Services, and private sector partners, Uganda

- Researched vanilla maturity and harvest timing for national harvest-declaration considerations, combining field and crop work with statistical modelling.
- Integrated field, satellite climate, and survey data and built time-series models to compare maturity-assessment methods and their timing, quality, and price implications.
- Applied thermal-time analysis based on accumulated growing degree days to link heat units to crop development through maturity.

- Conducted farmer and stakeholder interviews, field measurements of development stages, and seasonal maturity assessments across vanilla-growing districts.
- Helped establish the Kabanyolo demonstration plot and prepared models, scenario notes, and technical inputs for supervisors and stakeholders. Work concluded with program funding closure.

Research Associate, February 2023 - November 2023

Innovations for Poverty Action (IPA), Kampala, Uganda

- Supported development and implementation of process evaluation plans with program managers.
- Designed survey instruments and supported piloting, translation, and computer-assisted interviewing.
- Coordinated evaluation logistics including timelines, field staffing, and procurement, and managed data entry, cleaning, and analysis.
- Applied data-quality protocols including enumerator audits and monitoring of data collection.
- Maintained partner relationships, contributed to stakeholder meetings and reports, and supported research approvals and compliance.

Qualitative Research Associate, July 2023 - March 2024

Social and Scientific Systems for USAID Strategic Information Technical Support (SITES), Uganda

- Trained on field methods and tools and managed questionnaires and tablets for fieldwork across Eastern and Northern Uganda.
- Assessed facility readiness for Health Information Exchange and paperless viral load requests, reviewing UgandaEMR records in HIV clinics against ART registers, cards, and viral load registers.
- Conducted key informant interviews and focus group discussions with data clerks, data officers, clinicians, counselors, and laboratory staff, and checked data quality across HMIS tools and the EMR.
- Documented practical constraints affecting EMR use such as DHIS2 knowledge, internet and power stability, and computer performance, and shared facility-level feedback at each visit.
- Compiled progress and activity reports with safe back-up, storage, and transmission, and completed a 12-day assessment from 18 to 29 September 2023 within a seven-region exercise covering Kigezi, Ankole, Lango, Acholi, Busoga, Bugisu, and Karamoja.

Research Associate, September 2021 - January 2023

Bronkar (U) Limited, Kampala, Uganda

- Developed assignment workplans and tracked deliverables across research assignments.
- Designed methodologies and survey tools for qualitative and quantitative field collection.
- Trained enumerators and participated in field research including interviews and structured surveys.
- Cleaned and analyzed survey data and prepared insights and conclusions for client reports and proposals.

PORTFOLIO

- GitHub and portfolio: github.com/mugarrajohnson and johnsonmugarra.com - repositories and contribution history, project write-ups, live D3.js analytics, publications with DOIs, and a downloadable CV.
- Vanilla forecasting: Prophet and ARIMA models combining NASA POWER climate data with field measurements; interactive harvest model on the website, presented nationally at Makerere in April 2025.
- Farm data platform (Enimiro): seven scheduled Python and R pipelines moving SurveyCTO submissions into PostgreSQL, feeding Power BI and Looker dashboards; private, walk-through on request.
- Automated evaluation pipeline: Python ingestion, cleaning, and report generation turning recurring evaluation cycles into one rerunnable run.

EDUCATION

M.Sc. Data Science, September 2022 - October 2024

Manipal Academy of Higher Education (MAHE), Karnataka, India

B.Sc. Agricultural Science, September 2016 - May 2021

Makerere University, Kampala, Uganda

CERTIFICATIONS

- SAS Certified Statistical Business Analyst
- Microsoft Certified: Power BI Data Analyst Associate
- Tableau Desktop Specialist
- Google Project Management Certificate
- Human Subjects Research (CITI)

REFERENCE

- Available on request.